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(54) **WINDOW GLASS FOR VEHICLE**

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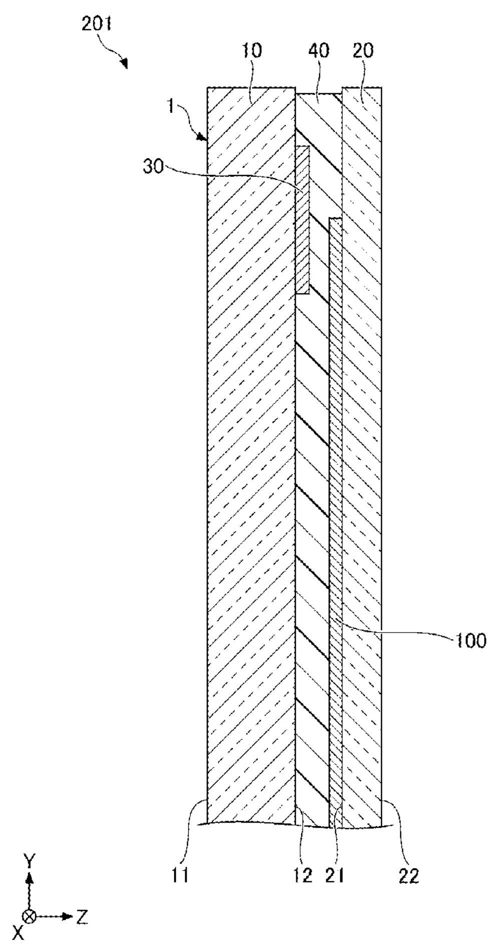
(57) **ABSTRACT**

A window glass for a vehicle comprises a laminated glass for a vehicle including a first glass plate having a first main surface and a second main surface opposite to the first main surface, a second glass plate having a third main surface facing the second main surface and a fourth main surface opposite to the third main surface, and an interlayer film disposed between the second main surface and the third main surface, a planar antenna disposed between the second main surface and the third main surface, and a conductive member that is separated from the antenna in a direction from the first glass plate toward the second glass plate and overlaps at least a part of the antenna in a plan view of the laminated glass.



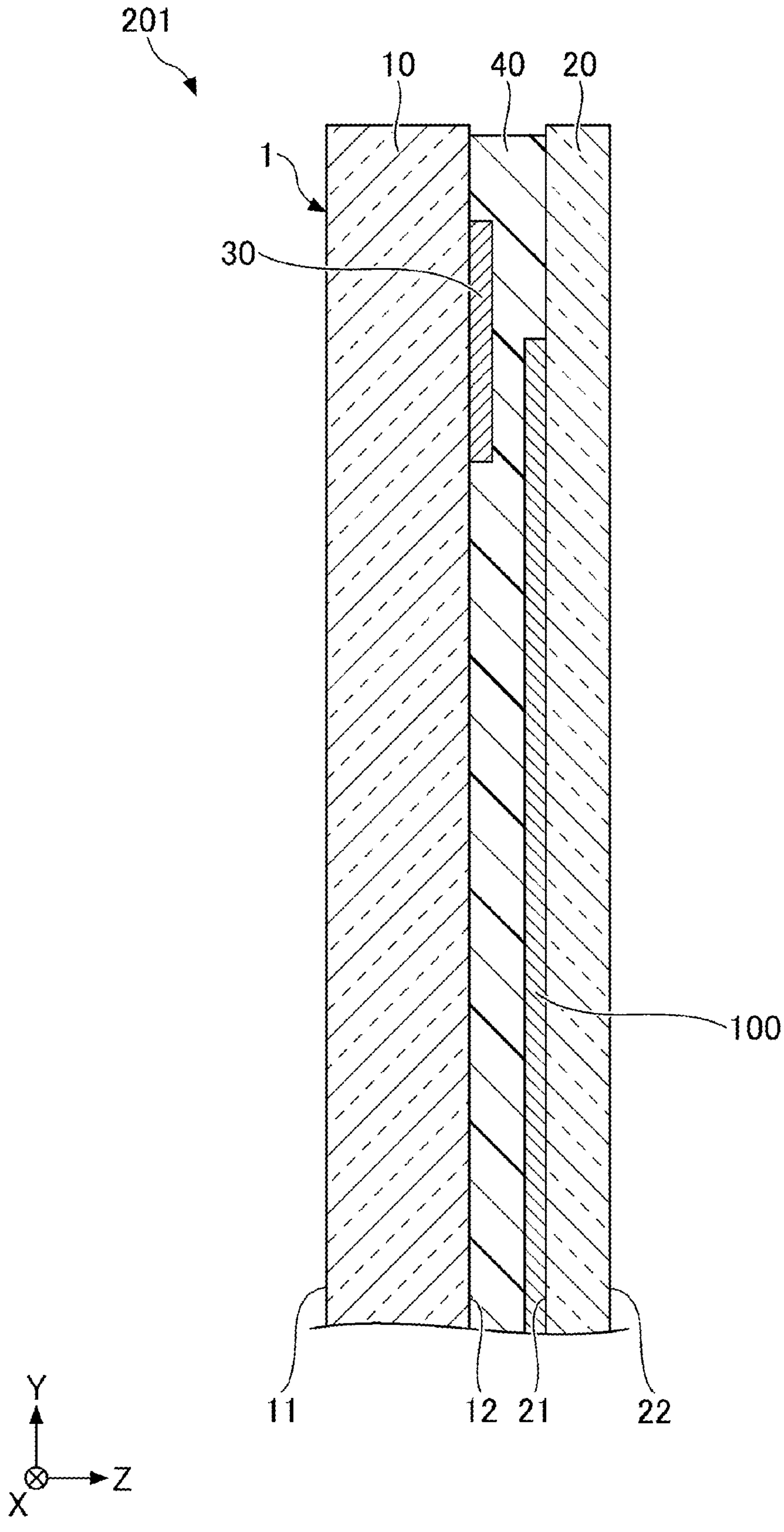


Fig. 1

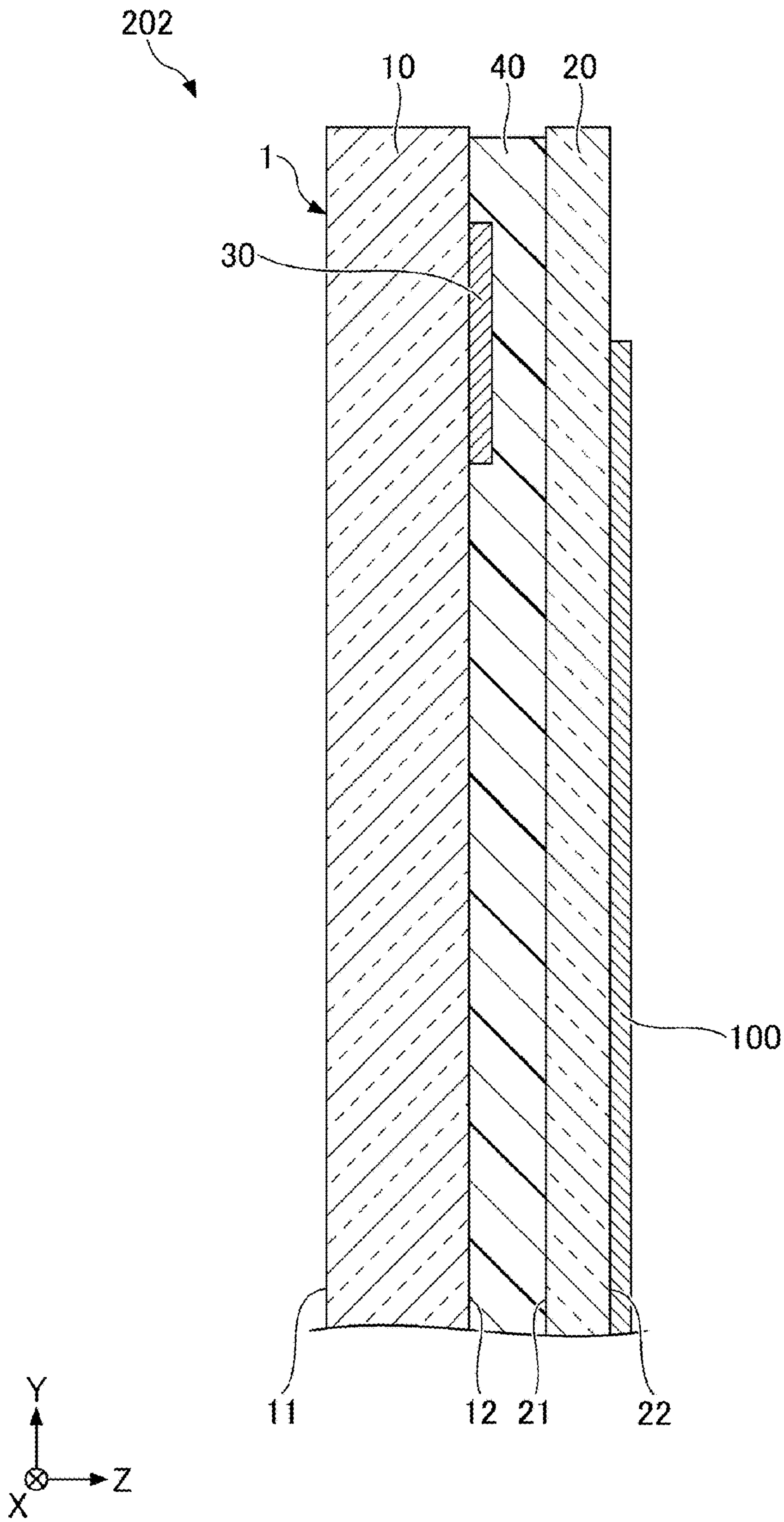


Fig. 2

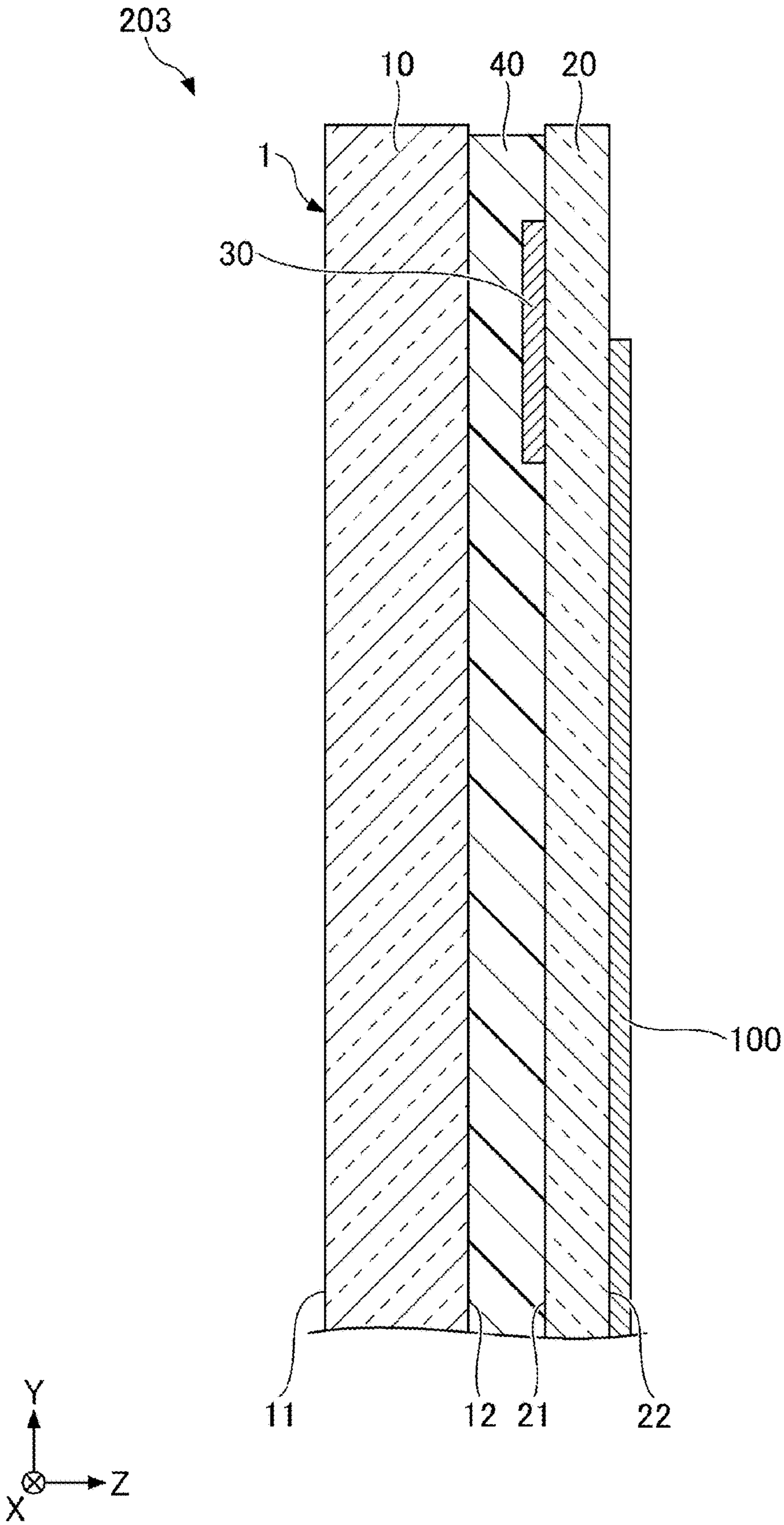


Fig. 3

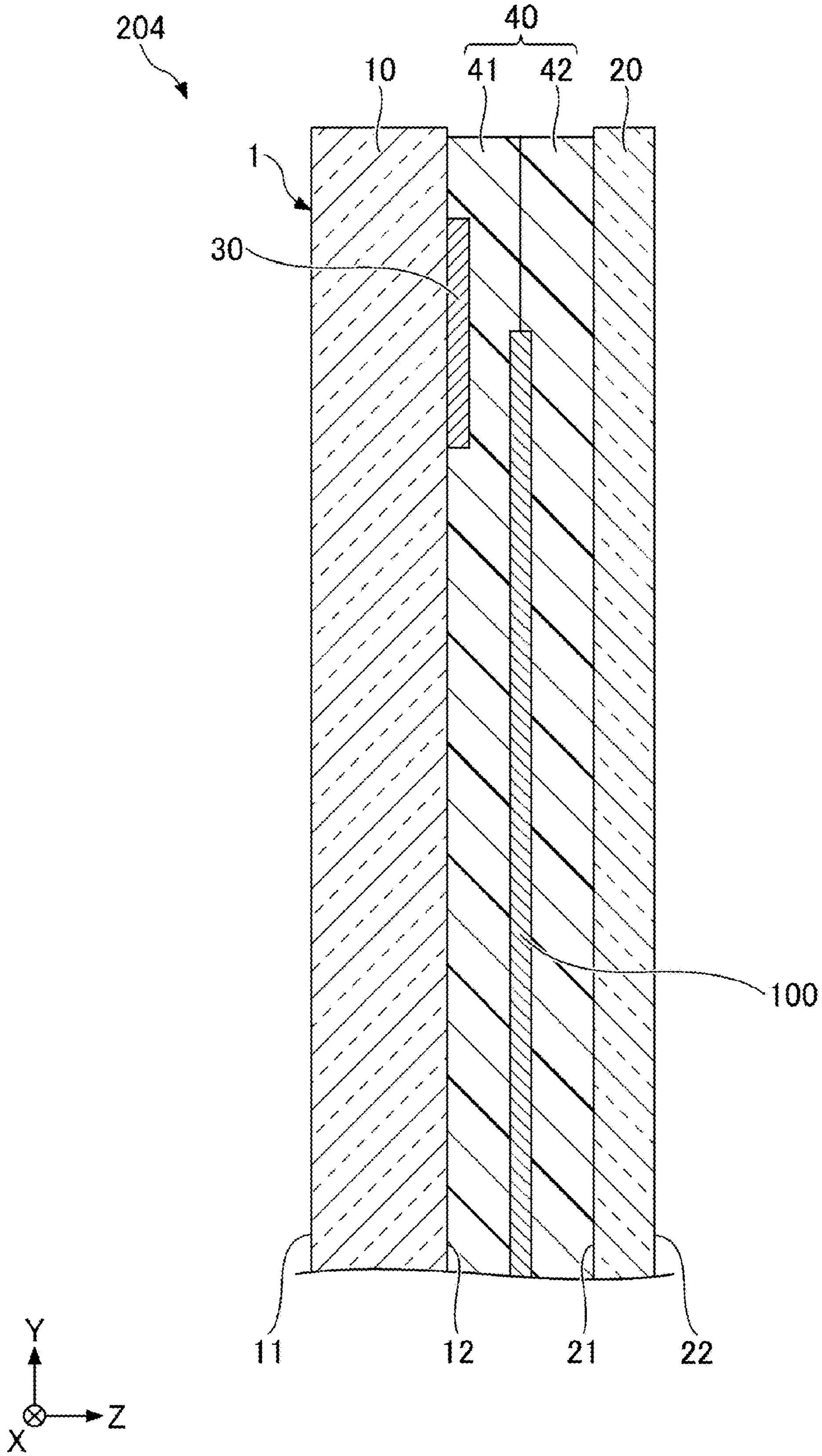


Fig. 4

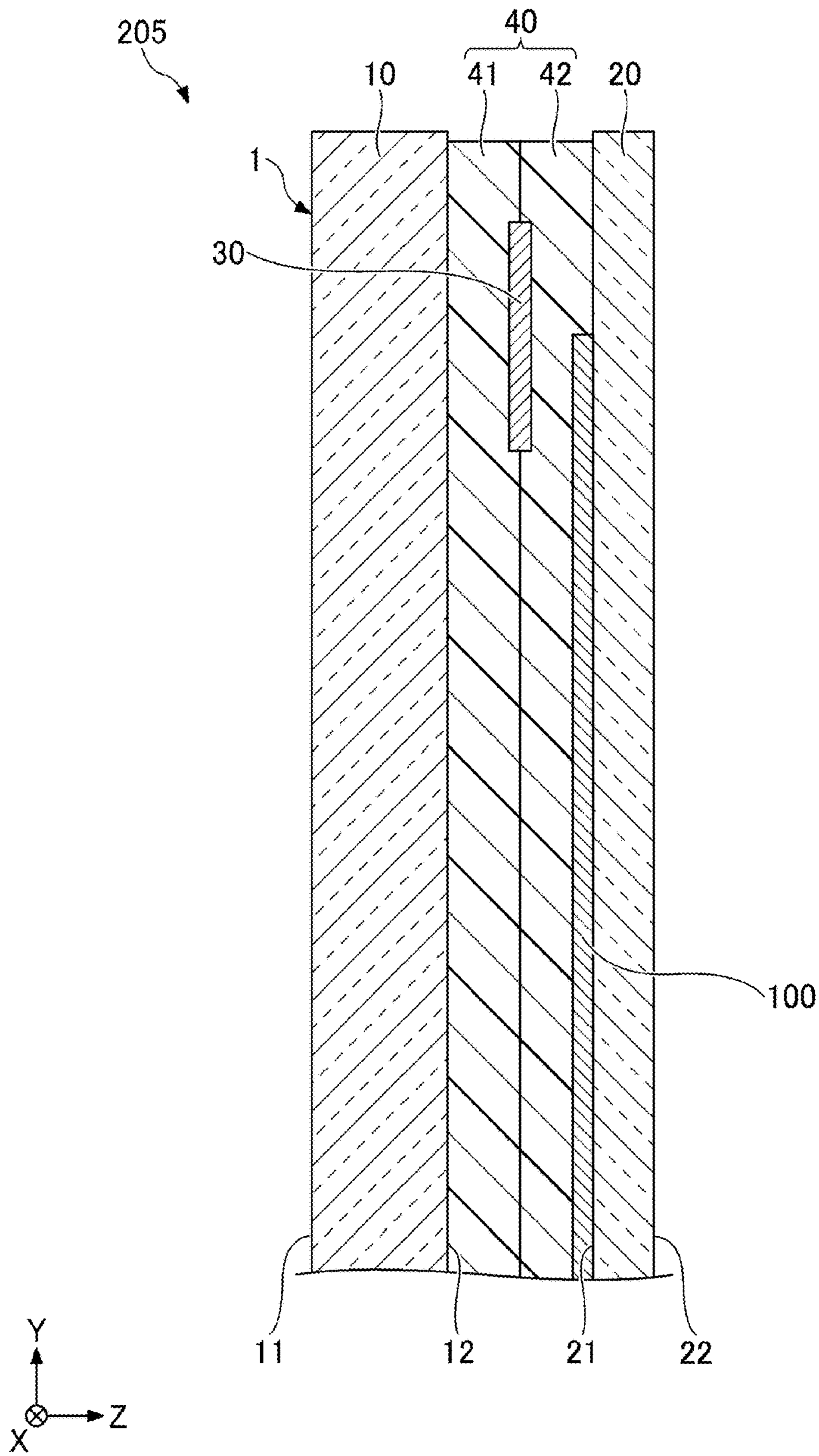


Fig. 5

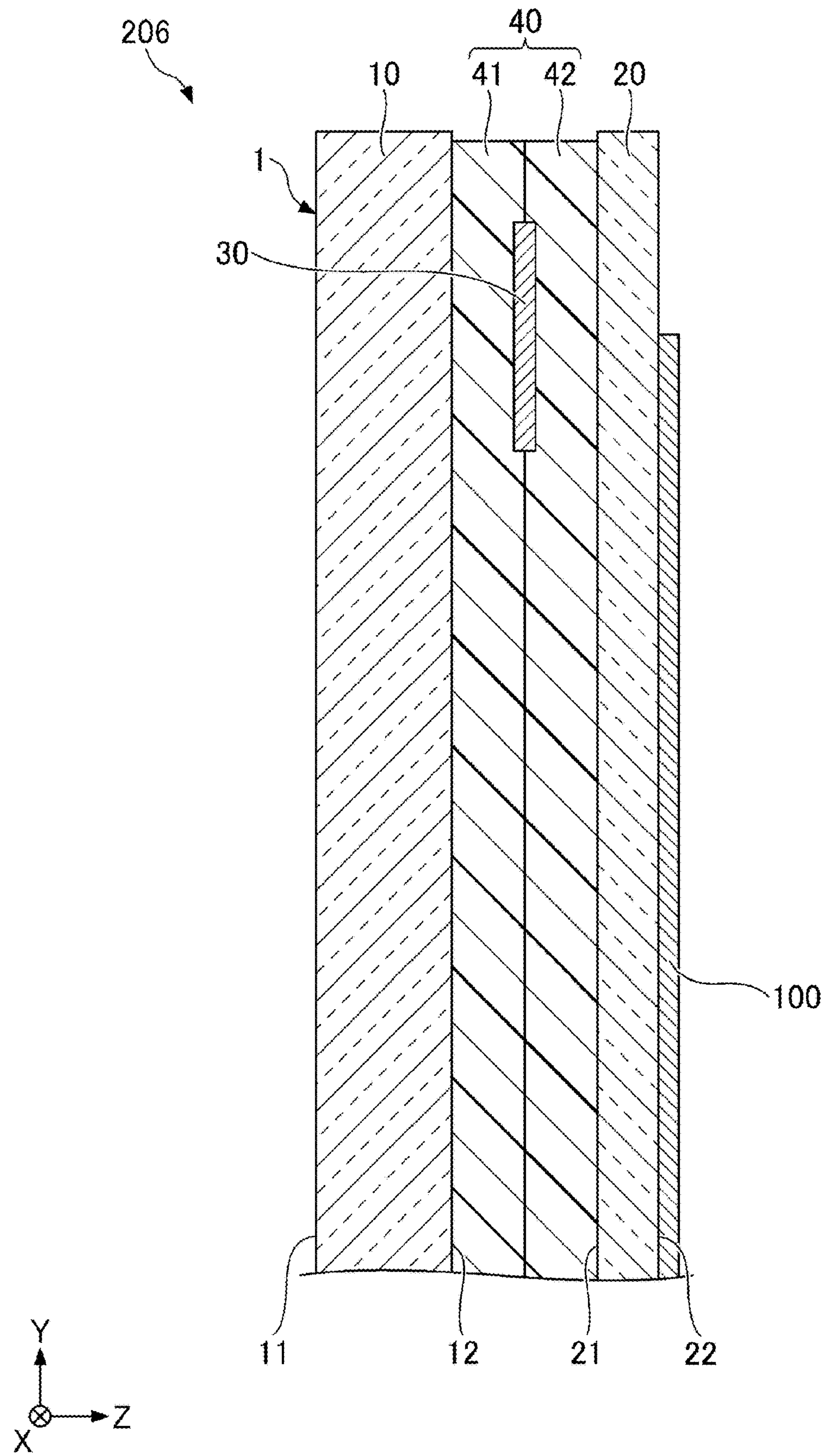


Fig. 6

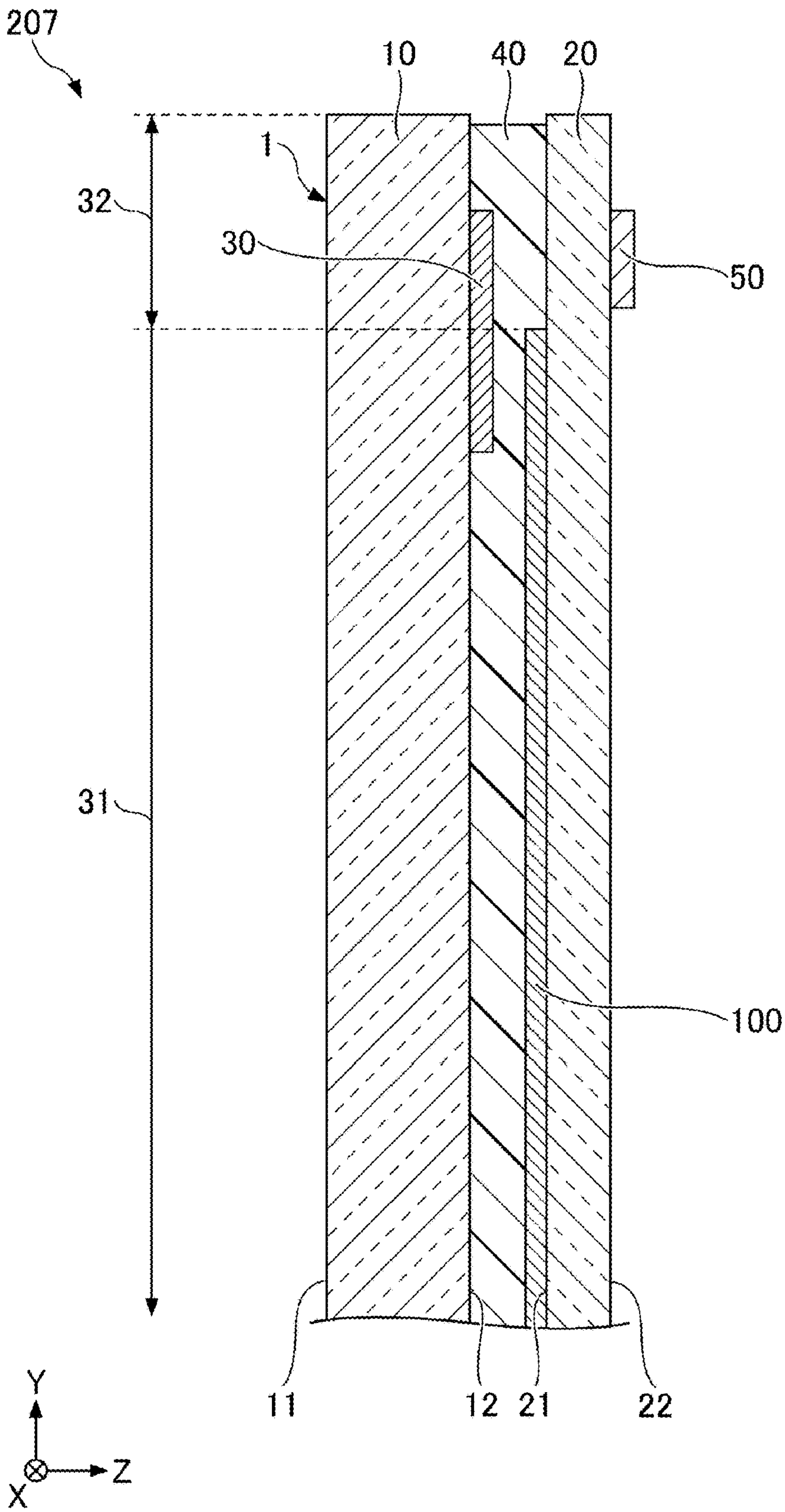


Fig. 7

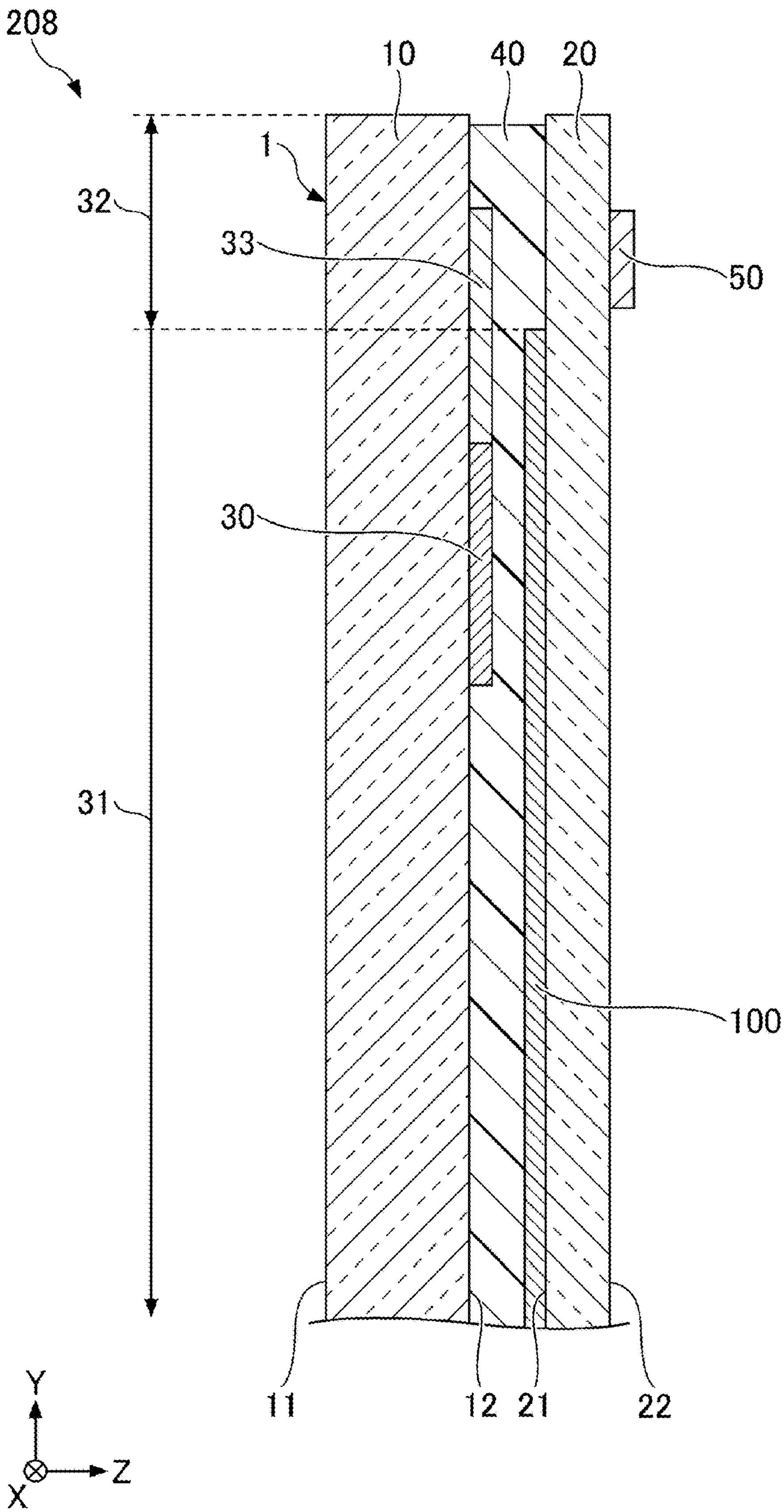


Fig. 8

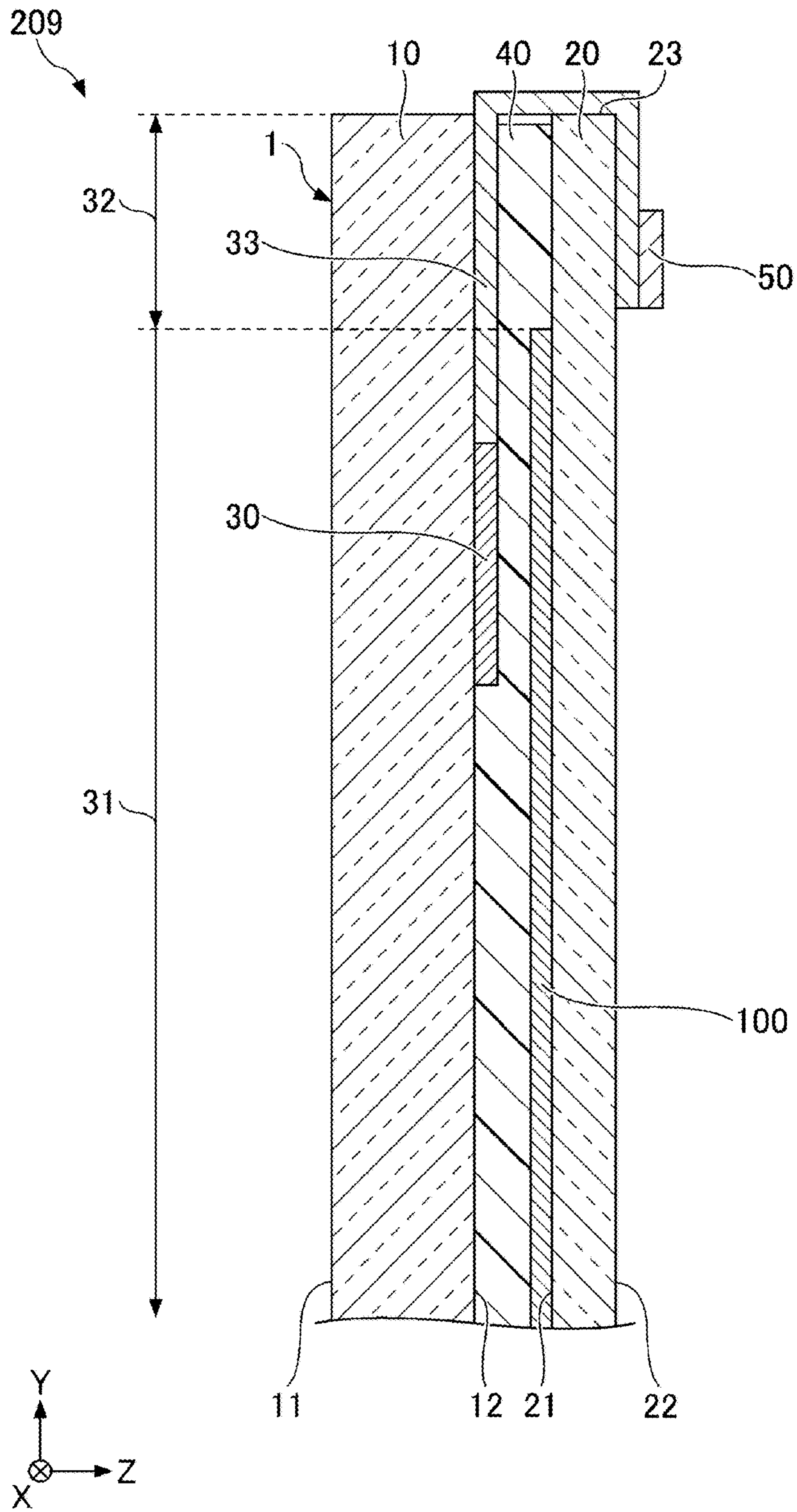


Fig. 9

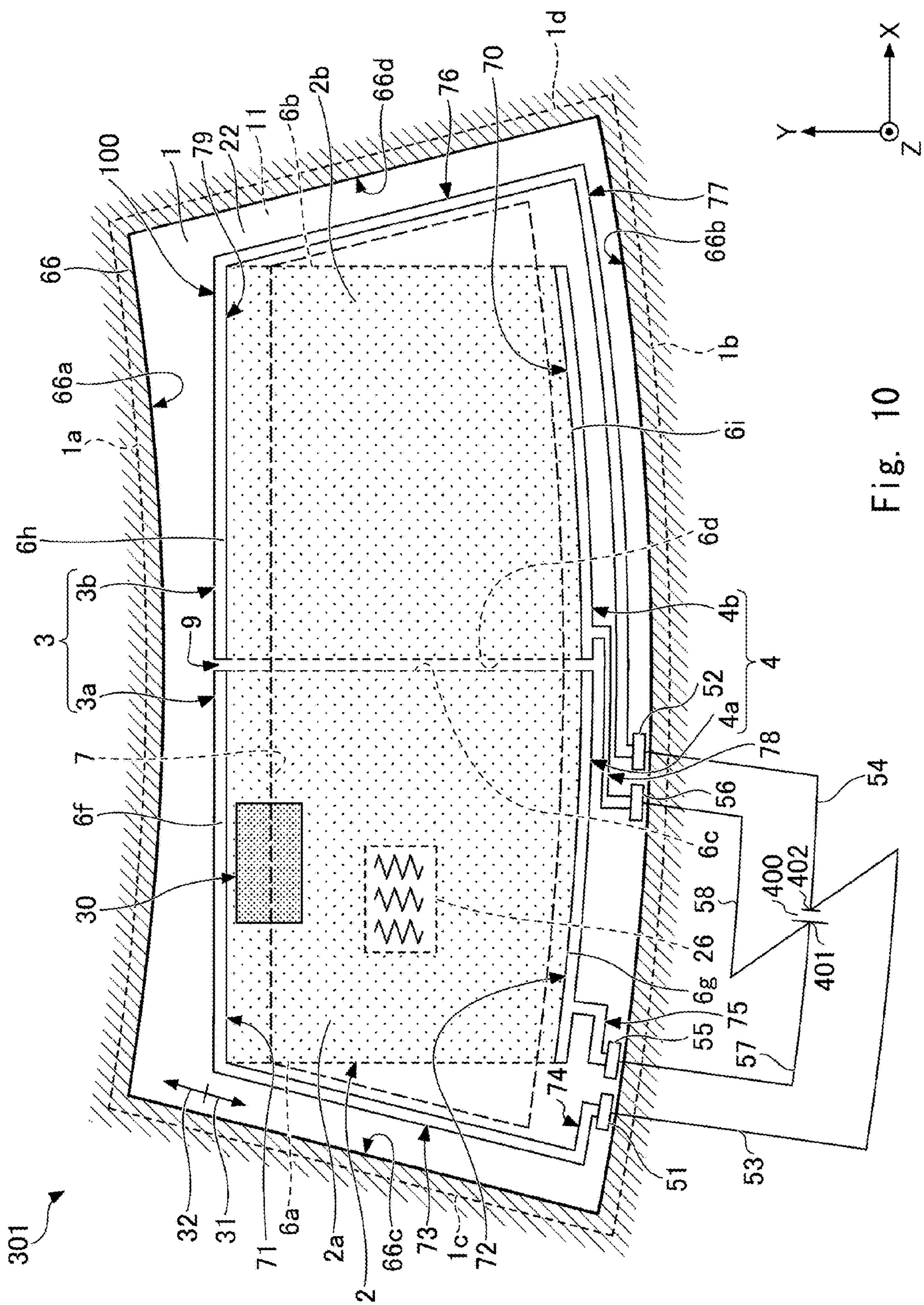


Fig. 10

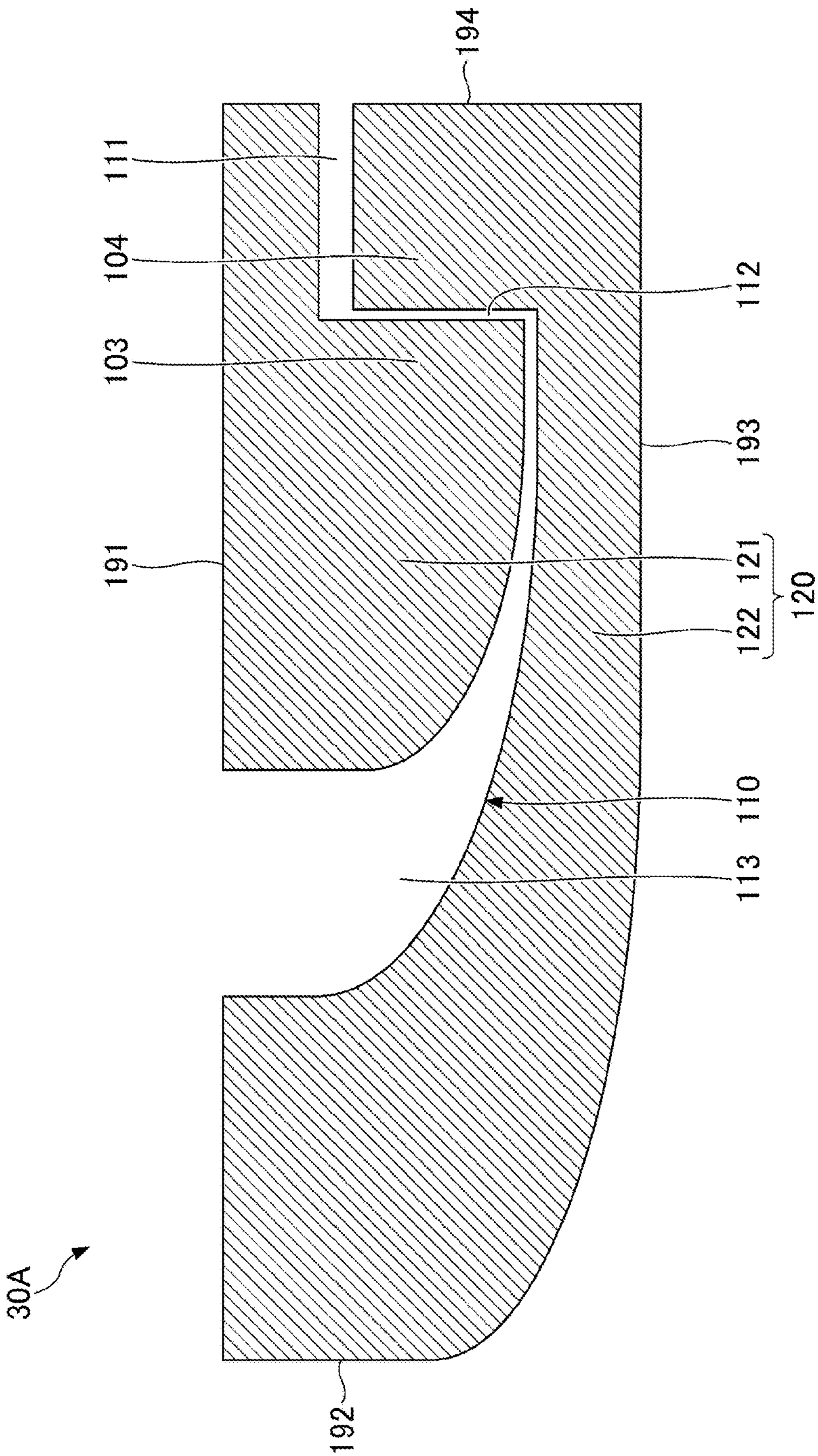


Fig. 11

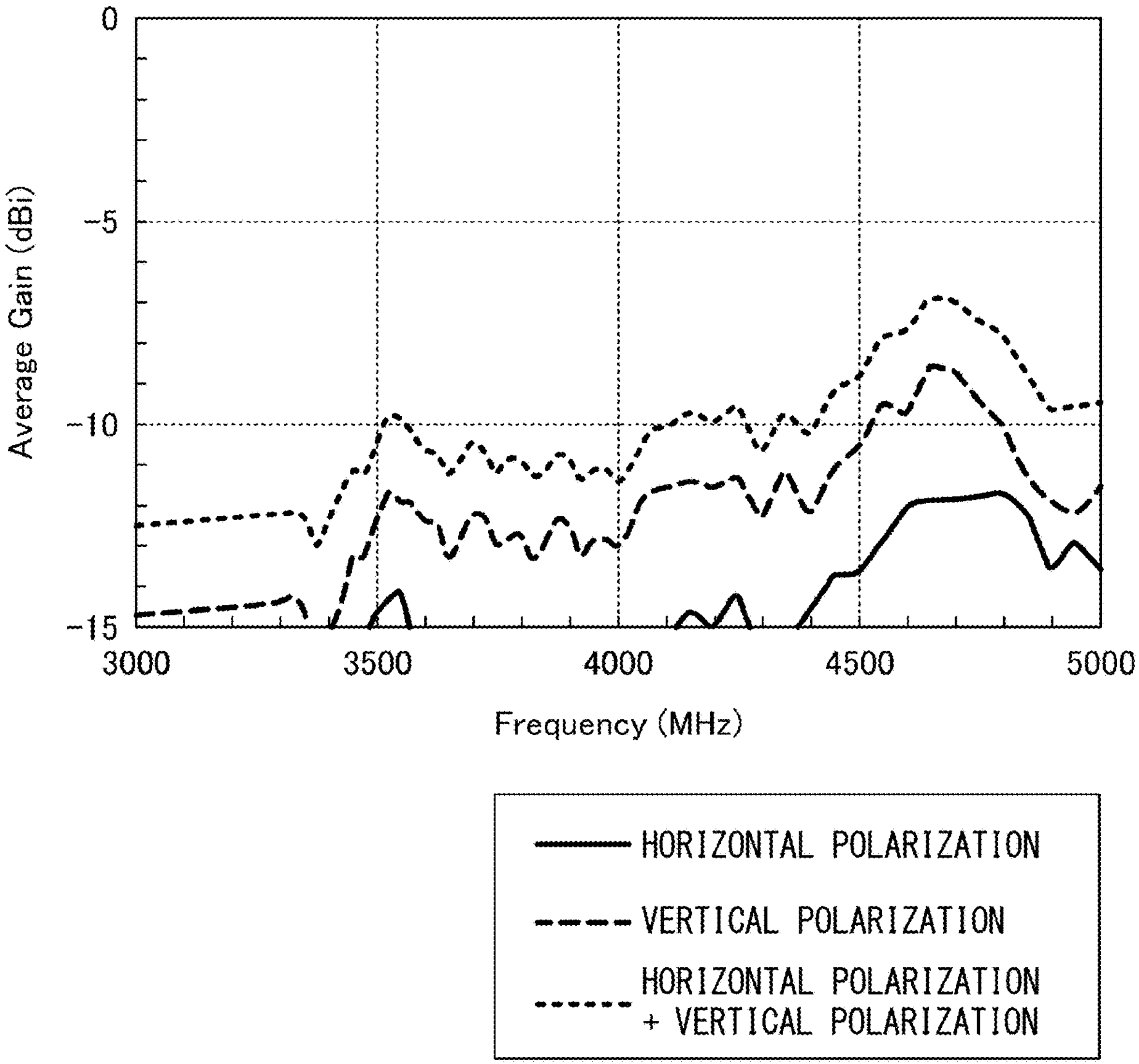

0
−5
−10
−15
Average Gain (dBi)
3000
3500
4000
4500
5000
Frequency (MHz)
HORIZONTAL POLARIZATION
VERTICAL POLARIZATION
HORIZONTAL POLARIZATION
+ VERTICAL POLARIZATION

Fig. 12

WINDOW GLASS FOR VEHICLE

INCORPORATION BY REFERENCE

[0001] This application is based upon and claims the benefit of priority from Japanese patent application No. 2022-186874, filed on Nov. 22, 2022, and PCT application No. PCT/JP2023/040907 filed on Nov. 14, 2023, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

[0002] Conventionally, there is known a window glass for a vehicle including a conductive member that heats a glass plate by applying a voltage to a pair of bus bars for anti-fogging and anti-icing and an antenna provided in the vicinity of a heating region where the conductive member is disposed (see, for example, International Patent Publication No. WO 2016/185898 and International Patent Publication No. WO 2016/096432).

SUMMARY

[0003] However, in a window glass in which an antenna is provided in a non-conductive region extending around a conductive region where a conductive member for anti-fogging or the like is disposed, if the degree of freedom of the shape and arrangement of the antenna is to be secured, an appropriate conductive region may not be secured. Conversely, when an attempt is made to secure a sufficient conductive region, the region where the antenna is disposed may be limited. As described above, there may be a trade-off relationship between the arrangement of the antenna and the arrangement of the conductive member.

[0004] The present disclosure provides a window glass for a vehicle in which it is easy to secure a region where an antenna and a conductive member are disposed.

[0005] The present disclosure provides a window glass for a vehicle including

[0006] a laminated glass for a vehicle including a first glass plate having a first main surface and a second main surface opposite to the first main surface, a second glass plate having a third main surface facing the second main surface and a fourth main surface opposite to the third main surface, and an interlayer film disposed between the second main surface and the third main surface,

[0007] a planar antenna disposed between the second main surface and the third main surface, and

[0008] a conductive member that is separated from the antenna in a direction from the first glass plate toward the second glass plate and overlaps at least a part of the antenna in a plan view of the laminated glass.

[0009] According to the present disclosure, it is possible to provide a window glass for a vehicle in which it is easy to secure a region where an antenna and a conductive member are disposed.

[0010] The above and other objects, features and advantages of the present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a first embodiment;

[0012] FIG. **2** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a second embodiment;

[0013] FIG. **3** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a third embodiment;

[0014] FIG. **4** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a fourth embodiment;

[0015] FIG. **5** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a fifth embodiment;

[0016] FIG. **6** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a sixth embodiment;

[0017] FIG. **7** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a seventh embodiment;

[0018] FIG. **8** is a cross-sectional view of an upper side portion of a window glass for a vehicle in an eighth embodiment;

[0019] FIG. **9** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a ninth embodiment;

[0020] FIG. **10** is a plan view illustrating a specific example of a window glass for a vehicle according to the present embodiment;

[0021] FIG. **11** is a plan view illustrating a specific example of a planar antenna according to the present embodiment; and

[0022] FIG. **12** is a diagram illustrating an example of a measurement result of frequency characteristics of an antenna.

DESCRIPTION OF EMBODIMENTS

[0023] Hereinafter, embodiments will be described with reference to the drawings. For easy understanding, the scale of each part in the drawings may be different from the actual scale. In directions such as parallel, right angle, orthogonal, horizontal, vertical, up and down, and left and right, and terms such as the same and equal, deviations are allowed to an extent that do not impair functions and effects of the embodiments. The shape of the corner portions is not limited to a right angle, and may be rounded in an arch shape. The term "facing" is not limited to a form in which all of them face each other, and may include a form in which some of them face each other. The "overlapping" is not limited to a form in which all of them overlap, and may include a form in which some of them overlap.

[0024] The X-axis direction, the Y-axis direction, and the Z-axis direction represent a direction parallel to the X-axis, a direction parallel to the Y-axis, and a direction parallel to the Z-axis, respectively. The X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to each other. The XY plane, the YZ plane, and the ZX plane represent a virtual plane parallel to the X-axis direction and the Y-axis direction, a virtual plane parallel to the Y-axis direction and the Z-axis direction, and a virtual plane parallel to the Z-axis direction and the X-axis direction, respectively. In the present embodiment, the X-axis direction is parallel to the left-right direction (lateral direction) of the vehicle body, the vehicle width direction of the vehicle body, or the horizontal direction (direction parallel to the horizontal plane).

[0025] As an example of the window glass for a vehicle in the present embodiment, a windshield attached to a front portion of a vehicle is preferable. However, the window glass for a vehicle is not limited to the windshield, and may be, for example, a rear glass attached to a rear portion of the vehicle, a side glass attached to a side portion of the vehicle, a roof glass attached to a ceiling portion of the vehicle, or the like. The window glass for a vehicle may be a window glass in which a roof glass is integrated with one or both of a windshield and a rear glass.

[0026] FIG. **1** is a cross-sectional view of an upper side portion when a window glass for a vehicle (windshield) is attached to a vehicle in a first embodiment. The positive side in the Z-axis direction corresponds to the vehicle inner side, and the negative side in the Z-axis direction corresponds to the vehicle outer side. A window glass for a vehicle **201** includes a glass plate **1**, an antenna **30**, and a conductive member **100**.

[0027] The glass plate **1** is an example of a laminated glass for vehicles. The glass plate **1** includes a glass plate **10**, a glass plate **20**, and an interlayer film **40**. In the example illustrated in FIG. **1**, the glass plate **1** is a laminated glass in which the glass plate **10** disposed on the vehicle outer side and the glass plate **20** disposed on the vehicle inner side are bonded to each other via the interlayer film **40**. The interlayer film **40** is sandwiched between the glass plate **10** and the glass plate **20**.

[0028] The glass plate **10** and the glass plate **20** are transparent dielectric plates. One or both of the glass plate **10** and the glass plate **20** may be translucent. The glass plate **10** is an example of a first glass plate, and the glass plate **20** is an example of a second glass plate.

[0029] The glass plate **10** has a main surface **11** facing a negative side in the Z-axis direction and a main surface **12** facing an opposite side (positive side in the Z-axis direction) to the main surface **11** in the Z-axis direction. The main surface **11** represents a surface on the vehicle outer side, and the main surface **12** represents a surface on the vehicle inner side. In particular, the main surface **11** corresponds to a surface of the laminated glass on the vehicle outer side. The main surface **11** is an example of a first main surface. The main surface **12** is an example of a second main surface.

[0030] The glass plate **20** is a dielectric plate facing the glass plate **10**. The glass plate **20** is disposed on the main surface **12** side with respect to the glass plate **10**. The glass plate **10** has a main surface **21** facing the main surface **12** of glass plate **10**, and a main surface **22** facing a side opposite to the main surface **21** in the Z-axis direction. The main surface **21** represents a surface on the vehicle outer side, and the main surface **22** represents a surface on the vehicle inner side. In particular, the main surface **22** corresponds to a surface of the laminated glass on the vehicle inner side. The main surface **21** is an example of a third main surface. The main surface **22** is an example of a fourth main surface.

[0031] In the case of the laminated glass, the thickness of the glass plate **10** is not particularly limited, but can be appropriately selected in the range of 0.1 mm to 10 mm. The thickness of the glass plate **10** is preferably 0.3 mm or more, more preferably 0.5 mm or more, still more preferably 0.7 mm or more, particularly preferably 1.1 mm or more, and most preferably 1.6 mm or more. In addition, the thickness of the glass plate **10** is preferably 3.0 mm or less, more preferably 2.6 mm or less, and still more preferably 2.1 mm or less so that the mass of the laminated glass does not become too large. The thickness of the glass plate **10** may be the same as or different from the thickness of the glass plate **20**. As long as the glass plate **10** and the glass plate **20** have the same thickness, glass plates having the same size can be used.

[0032] The interlayer film **40** is a film having dielectric properties and disposed between the main surface **12** of the glass plate **10** and the main surface **21** of the glass plate **20**. The interlayer film **40** is a transparent or translucent dielectric interposed between the glass plate **10** and the glass plate **20**. The glass plate **10** and the glass plate **20** are bonded by the interlayer film **40**. Examples of the interlayer film **40** include thermoplastic polyvinyl butyral (PVB), an ethylene vinyl acetate copolymer (EVA), and a cycloolefin polymer (COP). The relative permittivity of the interlayer film **40** is preferably 2.4 or more and 3.5 or less.

[0033] The antenna **30** is a planar antenna disposed between the main surface **12** and the main surface **21**. The antenna **30** may be a grid-like planar antenna in which at least one hole is formed in a planar conductor, a mesh-like planar antenna in which holes are finer and conductors are thinner, or a solid planar antenna in which holes are not formed in a planar conductor.

[0034] The antenna **30** is formed to be able to transmit and receive a radio wave in a predetermined frequency band (at least one of transmission and reception). The radio wave in the predetermined frequency band may be a vertically polarized wave, a horizontally polarized wave, or a circularly polarized wave. The predetermined frequency band is a relatively high frequency band (600 MHz to 3 GHZ) in an ultra high frequency (UHF) band, a super high frequency (SHF) band of 3 GHz to 30 GHz, or an extremely high frequency (EHF) band of 30 GHz to 300 GHz. As a specific example of such a high frequency band, there is a band (frequency band of 6 GHz or less (sub6) and frequency band of 24 GHz or more (28 GHz band, 39 GHz band, and the like)) used in the fifth generation communication (5G) standard.

[0035] The antenna **30** may be impedance matched to efficiently transmit and receive a radio wave of Wi-Fi, which is a wireless local area network (LAN). The antenna **30** may be impedance matched to transmit and receive radio waves in a frequency band (863 MHz to 868 MHz (Europe), 902 MHz to 928 MHz (USA), 2400 MHz to 2497 MHz (universal), 5150 MHz to 5350 MHZ (universal), 5470 MHz to 5850 MHz (universal), and the like) defined by the communication standards IEEE 802.11a, b, g, n, ac, ah, and ax.

[0036] The antenna **30** may be impedance matched to transmit and receive radio waves having a frequency of 2400 MHZ to 2483.5 MHz used in Bluetooth (registered trademark). The antenna **30** may be impedance matched to transmit and receive radio waves in a frequency band (755.5 MHz to 764.5 MHZ (Japan) defined by ARIB STD-T109, 5850 MHz to 5925 MHz defined by IEEE802.11p, and the like) used in vehicle-to-infrastructure (V2I) or vehicle-to-vehicle (V2V) communication in intelligent transport systems (ITS). The antenna **30** may be impedance matched to transmit and receive radio waves in a frequency band (2300 MHz to 2400 MHZ, 2496 MHz to 2690 MHz, 3400 MHz to

3600 MHZ, and the like) used in WiMAX (registered trademark) which is another wireless communication technology. The antenna **30** may be impedance matched to transmit and receive a radio wave in a low band (3245 MHz to 4742 MHZ) of an ultra wideband (UWB) wireless communication system.

[0037] The conductive member **100** is separated from the antenna **30** in a direction from the glass plate **10** toward the glass plate **20** (in this example, the positive Z-axis direction). The conductive member **100** is, for example, a member capable of generating heat for antifogging and the like. The conductive member **100** may have a function other than heat generation as long as it is a member having conductivity.

[0038] The conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1** in the positive Z-axis direction. Since the conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1**, it is easy to secure a region where the antenna **30** and the conductive member **100** are disposed as compared with a form in which the conductive member **100** does not overlap the antenna **30** in a plan view of the glass plate **1**. For example, even if the size required for the antenna **30** is secured, the region where the conductive member **100** is disposed can be sufficiently secured, and conversely, even if the size required for the antenna is secured in the region where the conductive member **100** is disposed, the size required for the antenna **30** can be sufficiently secured.

[0039] In addition, since the conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1**, the antenna **30** can use the conductive member **100** as a reflector that reflects radio waves, and thus the antenna gain (directivity) on the opposite side (in this example, the negative side in the Z-axis direction) to the side where the conductive member **100** is positioned with respect to the antenna **30** is improved.

[0040] The conductive member **100** may be a planar conductive film or a member including a plurality of heating wires **26** to be described later. Thus, the antenna can use the conductive member **100** as a reflector that reflects radio waves.

[0041] In a case where the conductive member **100** is a planar conductive film, when the sheet resistance of the conductive member (conductive film) is 5 [Ω/□] or more, in the antenna **30**, the conductive member **100** can be used as a reflecting member that reflects a radio wave in a frequency band transmitted and received by the antenna **30**, and the antenna gain (directivity) on the negative side in the Z-axis direction is improved. The sheet resistance is preferably 7 [Ω/□] or more, more preferably 10 [Ω/□] or more from the viewpoint of improving the antenna gain (directivity) on the negative side in the Z-axis direction. The upper limit value of the sheet resistance is not particularly limited, but may be 30 [Ω/□] or less.

[0042] In the example illustrated in FIG. **1**, the antenna **30** is disposed between the main surface **12** and the interlayer film **40**. The antenna **30** may be in contact with the main surface **12**, but a dielectric layer (not illustrated) such as a light shielding layer that shields visible light may be interposed between the antenna **30** and the main surface **12**. The light shielding layer (not illustrated) is, for example, an opaque colored ceramic layer, and the color is arbitrary, but dark color such as black, brown, gray, and deep blue or white is preferable, and black is more preferable.

[0043] In the example illustrated in FIG. **1**, the conductive member **100** is disposed between the interlayer film **40** and the main surface **21**. The conductive member **100** may be in contact with the main surface **21**, but a dielectric layer (not illustrated) such as a light shielding layer that shields visible light may be interposed between the conductive member **100** and the main surface **21**. Furthermore, a light shielding layer (not illustrated) may be disposed on the main surface **22**.

[0044] FIG. **2** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a second embodiment. In the second embodiment, the description of the same configuration, operation, and effect as those of the first embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **202** illustrated in FIG. **2** is different from the window glass for a vehicle **201** in the first embodiment in that the conductive member **100** is disposed on the main surface **22**. In the case of the second embodiment, since the conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1**, it is easy to secure a region where the antenna **30** and the conductive member **100** are disposed.

[0045] In FIG. **2**, the conductive member **100** may be in contact with the main surface **22**, but a dielectric layer (not illustrated) such as a light shielding layer that shields visible light may be interposed between the conductive member **100** and the main surface **22**. In a case where the conductive member **100** is disposed on the main surface **22**, the vehicle inner side which is the surface (the negative side in the Z-axis direction) thereof is preferably overcoated with a dielectric layer. In addition, the dielectric layer coating the conductive member **100** is preferably transparent in visible light.

[0046] FIG. **3** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a third embodiment. In the third embodiment, the description of the same configuration, operation, and effect as those of the above-described embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **203** illustrated in FIG. **3** is different from the window glass for a vehicle **202** in the second embodiment in that the antenna **30** is disposed on the main surface **21**. In the case of the third embodiment, since the conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1**, it is easy to secure a region where the antenna **30** and the conductive member **100** are disposed.

[0047] In FIG. **3**, the antenna **30** may be in contact with the main surface **21**, but a dielectric layer (not illustrated) such as a light shielding layer that shields visible light may be interposed between the antenna **30** and the main surface **21**. Furthermore, the light shielding layer (not illustrated) may be disposed in a part between the main surface **22** and the conductive member **100**.

[0048] FIG. **4** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a fourth embodiment. In the fourth embodiment, the description of the same configuration, operation, and effect as those of the above-described embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **204** illustrated in FIG. **4** is different from the window glass for a vehicle **201** in the first embodiment in that the interlayer film **40** includes an interlayer film **41** and an interlayer film **42** sandwiching the conductive member **100**

in a thickness direction (in this example, the Z-axis direction) of the glass plate **1**. In the case of the fourth embodiment, since the conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1**, it is easy to secure a region where the antenna **30** and the conductive member **100** are disposed.

[0049] As the description of the interlayer film **41** and the interlayer film **42**, the above description of the interlayer film **40** is incorporated. The interlayer film **41** is an example of a first interlayer film. The interlayer film **42** is an example of a second interlayer film. In this example, the conductive member **100** is sandwiched between the interlayer film **41** in contact with the main surface **12** and the interlayer film **42** in contact with the main surface **21**. A dielectric layer (not illustrated) such as a light shielding layer that shields visible light may be interposed between the interlayer film **41** and the main surface **12**, or may be interposed between the interlayer film **42** and the main surface **21**.

[0050] FIG. **5** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a fifth embodiment. In the fifth embodiment, the description of the same configuration, operation, and effect as those of the above-described embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **205** illustrated in FIG. **5** is different from the window glass for a vehicle **201** in the first embodiment in that the interlayer film **40** includes an interlayer film **41** and an interlayer film **42** sandwiching the antenna **30** in a thickness direction (in this example, the Z-axis direction) of the glass plate **1**. In the case of the fifth embodiment, since the conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1**, it is easy to secure a region where the antenna **30** and the conductive member **100** are disposed.

[0051] The interlayer film **42** is disposed between the interlayer film **41** and the main surface **21**. The conductive member **100** is disposed between the interlayer film **42** and the main surface **21**.

[0052] FIG. **6** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a sixth embodiment. In the sixth embodiment, the description of the same configuration, operation, and effect as those of the above-described embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **206** illustrated in FIG. **6** is different from the window glass for a vehicle **205** in the fifth embodiment in that the conductive member **100** is disposed on the main surface **22**. In the case of the sixth embodiment, since the conductive member **100** overlaps at least a part of the antenna **30** in a plan view of the glass plate **1**, it is easy to secure a region where the antenna **30** and the conductive member **100** are disposed.

[0053] FIG. **7** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a seventh embodiment. In the seventh embodiment, the description of the same configuration, operation, and effect as those of the above-described embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **207** illustrated in FIG. **7** is different from the window glass for a vehicle **201** in the first embodiment in that an electrode **50** provided on the main surface **22** is provided. The configuration described in the seventh embodiment may be applied to any of the plurality of embodiments described above.

[0054] The electrode **50** is connected to a transmission line (not illustrated) and is connected to a communication device (not illustrated) via the transmission line. Specific examples of the transmission line include a microstrip line, a strip line, a coplanar waveguide, a GCPW (coplanar waveguide with ground plane), a coplanar strip, a slot line, a waveguide, and the like. The transmission line may be a coaxial cable.

[0055] The electrode **50** is electrically connected to the antenna **30**, and in this example, is connected to the antenna **30** by capacitive coupling or electromagnetic coupling. The electrode **50** is capacitively coupled or electromagnetically coupled to the antenna **30** at an interval thinner than the thickness of the glass plate **10** in the Z-axis direction, for example. Since the electrode **50** and the antenna **30** are close to each other at a distance that can be capacitively coupled or electromagnetically coupled, the electrode **50** feeds signal to the antenna **30** sealed in the pair of glass plates **10** and **20** in a non-contact manner by capacitive coupling or electromagnetic coupling. By contactless feeding by capacitive coupling or electromagnetic coupling, even if the glass plate **20** and the interlayer film **40** are interposed between the antenna **30** and the electrode **50**, a simple feeding structure capable of feeding signal to the antenna **30** between the pair of glass plates **10** and can be realized. In addition, according to this feeding structure, even if the positive side end portion of the antenna **30** in the Y-axis direction is on the inner side (the negative side in the Y-axis direction) of the positive side end portion of the interlayer film **40** in the Y-axis direction, signal can be fed from the electrode **50** to the antenna **30** in the Z-axis direction with a simple structure.

[0056] In a plan view of the glass plate **1** in the positive Z-axis direction, a region where the conductive member **100** is disposed is referred to as a first region **31**, and a region positioned outside the first region **31** is referred to as a second region **32**. The electrode **50** is positioned in the second region **32** and faces the antenna **30** with the glass plate **20** interposed therebetween. With this arrangement, the electrode **50** can be connected to the antenna **30** by capacitive coupling or electromagnetic coupling even when the conductive member **100** is present. Note that, in FIG. **7**, the electrode **50** is illustrated as one electrode (in the drawings, a single electrode) for convenience, but a plurality of electrodes **50** (so-called bipolar) may be provided. That is, in a case where the electrode **50** is provided in a plurality of locations (for example, two), an electrode for a signal to be fed to the antenna and a ground electrode corresponding to the ground potential may be provided. In this case, when the transmission line (not illustrated) is a coaxial cable, a core wire of the coaxial cable is connected to the signal electrode of the electrode **50**, and a sheath wire is connected to the ground electrode.

[0057] The form in which the electrode **50** is provided on the main surface **22** may be a form in which the electrode **50** is provided on the bottom surface of the recess formed on the main surface **22**, a form in which a dielectric layer is interposed between the electrode **50** and the main surface **22**, a form in which a terminal provided with the electrode **50** is fitted into a hole formed in the main surface **22**, or the like. The hole formed in the main surface **22** may or may not penetrate from the main surface **22** to the main surface **21**.

[0058] The distance at which the antenna **30** and the electrode **50** can be electromagnetically coupled is, for example, 500 μm or less, preferably 250 μm or less, more preferably 150 μm or less, still more preferably 100 μm or less, and most preferably 50 μm or less.

[0059] The thickness of the glass plate **10** may be the same as the thickness of the glass plate **20**, or may be thicker than the glass plate **20**, for example. In a case where the thickness of the glass plate **10** is larger than the thickness of the glass plate **20**, the thickness of the glass plate **10** may be, for example, about 3.2 mm. In addition, the composition of the glass plate **20** can be appropriately selected, but a tempered glass is preferable as the glass plate capable of obtaining a predetermined strength with an electromagnetically couplable thickness as described above. Examples of the tempered glass include air-cooled tempered glass and chemically tempered glass, and as the tempered glass having a thin plate thickness, chemically tempered glass is preferable. In a case where the glass plate is chemically strengthened glass, the glass plate **20** may have a composition that can be strengthened by molding and chemical strengthening treatment. Examples of the glass plate that can be chemically strengthened include aluminosilicate glass, soda lime glass, borosilicate glass, lead glass, alkali barium glass, and aluminoborosilicate glass.

[0060] The feeding structure including the electrode **50** is suitable for feeding signal to the antenna **30** through which a high-frequency signal having a frequency in a relatively high band passes when signal is fed by contactless feeding by capacitive coupling or electromagnetic coupling. In addition, in order to reduce the loss of the high frequency signal, the dielectric loss tangent (tan δ) of the glass plate **20** is preferably low. For example, tan δ of the glass plate **20** at a frequency of 10 GHz is preferably 0.010 or less, and more preferably 0.009 or less.

[0061] FIG. **8** is a cross-sectional view of an upper side portion of a window glass for a vehicle in an eighth embodiment. In the eighth embodiment, the description of the same configuration, operation, and effect as those of the above-described embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **208** illustrated in FIG. **8** is different from the window glass for a vehicle **207** in the seventh embodiment in that a line **33** connected to the antenna **30** at the same layer is provided. The configuration described in the eighth embodiment may be applied to any of the plurality of embodiments described above.

[0062] The electrode **50** is electrically connected to the antenna **30**, and in this example, is connected to the line **33** connected in the same layer as the antenna **30** by capacitive coupling or electromagnetic coupling, and is electrically connected to the antenna **30** via the line **33**. The line **33** may include, for example, a transmission line as exemplified above. Specific examples of the line **33** include a flexible substrate on which a transmission line is formed. Since the electrode **50** and the line **33** are close to each other at a distance capable of being capacitively coupled or electromagnetically coupled, the electrode **50** feeds signal to the line **33** sealed in the pair of glass plates **10** and **20** in a non-contact manner by capacitive coupling or electromagnetic coupling, and can feed signal to the antenna **30** via the line **33**.

[0063] The electrode **50** is positioned in the second region **32** and faces the line **33** connected to the antenna **30** in the same layer with the glass plate **20** interposed therebetween. With this arrangement, the electrode **50** can be connected to the line **33** by capacitive coupling or electromagnetic coupling even when the conductive member **100** is present. In addition, since the line **33** is interposed, even if the entire antenna **30** overlaps the conductive member **100** in a plan view of the glass plate **1** (even if the antenna is included in the first region **31**), the electrode **50** can feed signal to the antenna **30** via the line **33**.

[0064] FIG. **9** is a cross-sectional view of an upper side portion of a window glass for a vehicle in a ninth embodiment. In the ninth embodiment, the description of the same configuration, operation, and effect as those of the above-described embodiment will be omitted or simplified by incorporating the above description. A window glass for a vehicle **209** illustrated in FIG. **9** is different from the window glass for a vehicle **208** in the eighth embodiment in that an electrode **50** is connected to an antenna **30** via a line **33** passing through the outside of an end surface **23** of a glass plate **20**. The configuration described in the ninth embodiment may be applied to any of the plurality of embodiments described above.

[0065] The electrode **50** is electrically connected to the antenna **30**, and in this example, is electrically connected to the antenna **30** via the line **33** passing through the outside of the end surface **23** of the glass plate **20**. The line **33** may include, for example, a transmission line as exemplified above. Specific examples of the line **33** include a flat harness in which a transmission line is formed. The electrode **50** can feed signal to the antenna **30** via the line **33**. In FIG. **9**, the electrode **50** is positioned in the second region **32**, but may be positioned in the first region **31** or may be positioned across the first region **31** and the second region **32**.

[0066] FIG. **10** is a plan view illustrating a specific example of a window glass for a vehicle according to the present embodiment. The plurality of cross-sectional structures described above may be applied to the configuration illustrated in FIG. **10**. FIG. **10** illustrates a window glass for a vehicle **301** attached to a window frame **66** of the vehicle body as viewed from inside the vehicle. The window glass for a vehicle **301** is, for example, a windshield attached to the window frame **66** formed at the front portion of the vehicle body.

[0067] The window frame **66** has an upper frame **66*a***, a lower frame **66*b***, a left frame **66*c***, and a right frame **66*d*** to form an opening covered by the window glass for a vehicle **301**. The upper frame **66*a*** is a window frame portion extending in the X-axis direction on the positive side in the Y-axis direction of the vehicle body, and is, for example, a flange on the ceiling side of the vehicle body. The lower frame **66*b*** is a window frame portion extending in the X-axis direction on the negative side in the Y-axis direction of the vehicle body, and is, for example, a flange on the dash panel side of the vehicle body. The left frame **66*c*** is a window frame portion connecting the upper frame **66*a*** and the lower frame **66*b*** on the negative side in the X-axis direction of the vehicle body, and is, for example, an A-pillar flange on the front left side of the vehicle body. The right frame **66*d*** is a window frame portion connecting the upper frame **66*a*** and the lower frame **66*b*** on the positive side in the X-axis direction of the vehicle body, and is, for example, an A-pillar flange on the front right side of the vehicle body.

[0068] The window glass for a vehicle **301** includes the glass plate **1**, the antenna **30**, and the conductive member **100**. In this example, the conductive member **100** includes a first bus bar **3**, a second bus bar **4**, and a heating element **2**. In FIG. **10**, the antenna **30** is disposed in a first heating region **2***a* to be described later, but is not limited thereto, and may be disposed in a second heating region **2***b* to be described later. Furthermore, the number of antennas **30** is not limited to one, and a plurality of antennas may be provided. In this case, the antennas may be disposed in any one of the first heating region **2***a* and the second heating region **2***b*, and one or more antennas may be disposed in each region.

[0069] The glass plate **1** is an example of a laminated glass for vehicles. The glass plate **1** is a transparent or translucent dielectric plate attached to the window frame **66**. The glass plate **1** has an outer peripheral edge including an upper edge **1***a*, a lower edge **1***b*, a left edge **1***c*, and a right edge **1***d*. The upper edge **1***a* is a glass edge extending in the X-axis direction on the positive side in the Y-axis direction of the vehicle body, and is attached to the upper frame **66***a*. The lower edge **1***b* is a glass edge extending in the X-axis direction on the negative side in the Y-axis direction of the vehicle body, and is attached to the lower frame **66***b*. The left edge **1***c* is a glass edge connecting the upper edge **1***a* and the lower edge **1***b* on the negative side in the X-axis direction of the vehicle body, and is attached to the left frame **66***c*. The right edge **1***d* is a glass edge connecting the upper edge **1***a* and the lower edge **1***b* on the positive side in the X-axis direction of the vehicle body, and is attached to the right frame **66***d*.

[0070] The glass plate **1** has the main surface **22** and the main surface **11** opposite to the main surface **22**. In this example, the main surface **22** is a surface on the vehicle inner side, and the main surface **11** is a surface on the vehicle outer side.

[0071] The first bus bar **3** is a strip-shaped electrode provided on the glass plate **1**. The first bus bar **3** includes upper portions **71** and **79** extending in a direction (for example, in a substantially horizontal direction,) along the upper edge **1***a* of the glass plate **1**. The first bus bar **3** is conductively connected to one electrode terminal (for example, the negative electrode terminal **402**) of a power source **400** mounted on the vehicle.

[0072] The second bus bar **4** is a strip-shaped electrode provided on the glass plate **1** and spaced apart from the first bus bar **3** on the negative side in the Y-axis direction. The second bus bar **4** includes lower portions **72** and **70** extending in a direction (for example, in a substantially horizontal direction) along the lower edge **1***b* of the glass plate **1**. The second bus bar **4** is conductively connected to the other electrode terminal (for example, a positive electrode terminal **401**) of the power source **400** mounted on the vehicle.

[0073] The first bus bar **3** may be conductively connected to a positive electrode terminal **401** of the power source **400**, and the second bus bar **4** may be conductively connected to the negative electrode terminal **402** of the power source **400**.

[0074] The heating element **2** is connected between the first bus bar **3** and the second bus bar **4**. The heating element **2** forms a heating region extending between the upper portions **71** and **79** and the lower portions **72** and **70**. The heating region is a region where the heating element **2** is disposed, and is heated by heat generated by the heating element **2**. The heating region has a pair of lateral sides (left side **6***a* and right side **6***b*) facing each other in the X-axis direction.

[0075] The heating element **2** is provided on the glass plate **1** and is positioned between the upper portions **71** and **79** and the lower portions **72** and **70**. The heating element **2** is a member through which a direct electrical current flows in the vertical direction between the upper portions **71** and **79** and the lower portions **72** and **70** when the direct voltage is applied between the first bus bar **3** and the second bus bar **4** by the power source **400**, and generates heat when the direct electrical current flows in the vertical direction. The heating region where the heating element **2** is disposed is heated by heat generated by the heating element **2** that conductively connects the upper portions **71** and **79** and the lower portions **72** and **70**. By heating the heating region, melting of snow, melting of ice, anti-fogging, and the like in the heating region and a region in the vicinity of the heating region in the glass plate **1** can be performed.

[0076] The heating element **2** is disposed on the same layer (inner layer or main surface **22**) as the first bus bar **3** and the second bus bar **4**. However, the heating element **2** may be disposed in a layer different from at least one of the first bus bar **3** and the second bus bar **4** as long as electrical connection with the first bus bar **3** and the second bus bar **4** is secured via the auxiliary member.

[0077] The heating region where the heating element **2** is disposed may be separated into a plurality of heating regions disposed in the X-axis direction. In this example, the heating region has two regions arranged in the X-axis direction via a gap **9** whose longitudinal direction is the vertical direction of the glass plate **1**, that is, the first heating region **2***a* and the second heating region **2***b*. The heating region where the heating element **2** is disposed may have three or more regions.

[0078] The first heating region **2***a* has a pair of lateral sides (upper side **6***f* and lower side **6***g*) facing each other in the Y-axis direction, and a pair of longitudinal sides (left side **6***a* and right side **6***c*) facing each other in the Y-axis direction. The heating element **2** disposed in the first heating region **2***a* is conductively connected to the upper portion **71** at an upper side **6***e*, and is conductively connected to the lower portion **72** at the lower side **6***g*.

[0079] The second heating region **2***b* has a pair of lateral sides (upper side **6***h* and lower side **6***i*) facing each other in the Y-axis direction, and a pair of longitudinal sides (left side **6***d* and right side **6***b*) facing each other in the Y-axis direction. The heating element **2** disposed in the second heating region **2***b* is conductively connected to the upper portion **79** at the upper side **6***h*, and is conductively connected to a lower portion **70** at the lower side **6***i*.

[0080] In this example, since the heating region where the heating element **2** is disposed is divided into the plurality of heating regions, the first bus bar **3** and the second bus bar **4** are also divided. The first bus bar **3** includes a first upper bus bar **3***a* and a second upper bus bar **3***b*, and the second bus bar **4** includes a first lower bus bar **4***a* and a second lower bus bar **4***b*.

[0081] The first bus bar **3** may further include a vertical portion connected to the upper portions **71** and **79**. In this example, in the first bus bar **3**, the first upper bus bar **3***a* includes a vertical portion **73** connected to the upper portion **71**, and the second upper bus bar **3***b* includes a vertical portion **76** connected to the upper portion **79**. The upper portion **71** is a conductor portion connected to the upper side **6***f* of the first heating region **2***a*, and the vertical portion **73** is a conductor portion extending in a direction along the left edge **1***c*, which is one side edge of the glass plate **1**, and is located away from the left side **6***a*, which is one side edge of the first heating region **2***a*. The upper portion **79** is a conductor portion connected to the upper side of the second heating region **2***b*, and the vertical portion **76** is a conductor portion extending in a direction along the right edge **1***d*, which is the other side edge of the glass plate **1**, and is located away from the right side **6***b*, which is one side edge of the second heating region **2***b*.

[0082] Since the first bus bar **3** includes the vertical portions **73** and **76** respectively connected to the upper portions **71** and **79**, a part of the wiring line electrically connecting the upper portions **71** and **79** of the first bus bar **3** to the power source **400** can be provided on the glass plate **1** side instead of the vehicle body side. Thus, the length of the harness wired to the vehicle body side can be reduced.

[0083] The first bus bar **3** may further include a lateral portion **74** connected to the vertical portion **73**, and may further include a lateral portion **77** connected to the vertical portion **76**. The lateral portion **74** is a conductor portion extending in a direction along the lower edge **1***b* of the glass plate **1** in a region away from the first heating region **2***a*. The lateral portion **77** is a conductor portion extending in a direction along the lower edge **1***b* of the glass plate **1** in a region away from the second heating region **2***b*. Due to the presence of the lateral portion **74** or the lateral portion **77**, the length of the harness can be further reduced depending on the position of the terminal of the harness to be wired to the vehicle body side.

[0084] In this example, the glass plate **1** includes a plurality of electrodes **51**, **52**, **55**, and **56** to which terminals of a plurality of harnesses electrically connected to the power source **400** are electrically connected.

[0085] The electrode **51** is a negative electrode for electrically connecting a terminal of a ground harness **53** electrically connected to the negative electrode terminal **402** to the first upper bus bar **3***a*. The electrode **51** is electrically connected to the upper portion **71** via the lateral portion **74** and the vertical portion **73**.

[0086] The electrode **52** is a negative electrode for electrically connecting a terminal of a ground harness **54** electrically connected to the negative electrode terminal **402** to the second upper bus bar **3***b*. The electrode **52** is electrically connected to the upper portion **79** via the lateral portion **77** and the vertical portion **76**.

[0087] The electrode **55** is a positive electrode for electrically connecting a terminal of a power supply harness **57** electrically connected to the positive electrode terminal **401** to the first lower bus bar **4***a*. The first lower bus bar **4***a* has a connection bus bar **75** connected to a lower portion **72**. The electrode **55** is electrically connected to the lower portion **72** via the connection bus bar **75**.

[0088] The electrode **56** is a positive electrode for electrically connecting a terminal of a power supply harness **58** electrically connected to the positive electrode terminal **401** to the second lower bus bar **4***b*. The second lower bus bar **4***b* has a connection bus bar **78** connected to the lower portion **70**. The electrode **56** is electrically connected to the lower portion **70** via the connection bus bar **78**.

[0089] As viewed in an enlarged manner in the drawings, for example, the heating element **2** includes a plurality of heating wires **26** extending in the vertical direction of the glass plate **1** and disposed at intervals in the X-axis direction. The plurality of heating wires **26** are connected between the upper portions **71** and **79** and the lower portions **72** and **70**. The plurality of heating wires **26** are, for example, wavy filament conductors extending from the first bus bar **3** toward the second bus bar **4**. The heating wire is formed of, for example, copper, aluminum, chromium, molybdenum, nickel, titanium, palladium, indium, tungsten, gold, platinum, silver, or an alloy containing a plurality of these elements.

[0090] The wire diameter of the heating wire **26** is, for example, 22 μm or more and 30 μm or less, or 22 μm or more and 25 μm or less. The interval (pitch) between adjacent heating wires **26** may be, for example, 2.0 mm or more and 3.0 mm or less, or 2.4 mm or more and 3.0 mm or less. The resistance of the heating wire **26** is, for example, 180 Ω/m at 20° C. when the wire diameter is 22 μm, and 138 Ω/m at 20° C. when the wire diameter is 25 μm.

[0091] An interval (pitch) between adjacent the heating wires **26** can be adjusted according to a frequency at which the antenna **30** performs transmission and reception. That is, in order to efficiently reflect the radio wave having a frequency F transmitted and received by the antenna **30** by the heating wire **26**, when a wavelength at a predetermined frequency F is λ, an interval P [mm] between the heating wires **26** adjacent to each other may be λ/25 or less. Note that A corresponds to a wavelength including a wavelength shortening rate of a (dielectric) medium such as glass encapsulating the antenna **30**, and when a wavelength in the air at a predetermined frequency F is λ_0 and a wavelength shortening rate of a peripheral medium encapsulating the antenna **30** is k, $\lambda=\lambda_0\times\lambda$ is satisfied. For example, in a case where the interval P between the heating wires **26** is 2.7 [mm], $\lambda\geq 67.5$ [mm], and a radio wave having a frequency of 4.44 [GHz] or more can be efficiently reflected by the heating wires **26**. Note that, in the above description, the heating wire **26** disposed at the predetermined interval P [mm] and having a width narrower than the interval P has been exemplified, but the conductive wire including the heating wire **26** does not necessarily have electric heating properties, and a parasitic conductive wire to which no voltage is applied may be disposed.

[0092] In addition, the conductive member **100** may be a transparent or translucent planar conductive film disposed on the inner layer or the main surface **22** of the glass plate **1**, a heat generating wire disposed on the inner layer or the surface of the glass plate **1**, or a silver-based print formed on the surface of the glass plate **1**. In a case where the conductive member **100** is installed on the inner layer of the glass plate **1**, the conductive member **100** is sealed in the laminated glass (glass plate **1**).

[0093] The conductive member **100** is, for example, a planar conductor disposed on the inner layer or the main surface **22** side of the glass plate **1**. The conductive member **100** may be a conductor that is in contact with the main surface **22** or a conductor that sandwiches the indirect member with the main surface **22**. Specific examples of the conductive member **100** include a metal film such as an Ag (silver) film, a metal oxide film such as an indium tin oxide (ITO) film, a resin film containing conductive fine particles, and a laminate in which a plurality of types of films are laminated. The conductive member **100** may be formed by coating a resin film such as polyethylene terephthalate by vapor deposition or the like. The conductive member **100** may be formed in a mesh shape on a film by conductive ink or etching.

[0094] The conductive member **100** may be a conductive film coated on the main surface **22** of the glass plate **1**. Specific examples of the conductive film include a low-radiation film such as a low emissivity (low-E) film exhibiting low radiation performance.

[0095] The low radiation means to reduce heat transfer due to radiation. A low-radiation film such as a low-E film suppresses heat transfer due to radiation, thereby securing heat insulating properties. The low-radiation film may be a general film, and may be, for example, a laminated film including a transparent dielectric film, an infrared reflecting film, and a transparent dielectric film in this order. The transparent dielectric material film is typically a metal oxide or a metal nitride. As the metal oxide, zinc oxide and tin oxide are representative. The infrared reflecting film is typically a metal film. The metal film is typically silver (Ag). Here, one or more infrared reflecting films may be formed between the transparent dielectric material films.

[0096] In addition, the conductive member **100** is not limited to a low-radiation film such as a low-E film, and may have other functions as long as it is a conductive layer. For example, the conductive member **100** may have functions such as anti-icing and anti-fogging of the window glass by heat generation due to voltage application.

[0097] Further, the conductive member **100** may be a conductive film included in a light control film capable of actively changing the visible light transmittance of the glass plate **1** by applying an AC voltage. The light control film has, for example, a molecular layer (not illustrated) having optical anisotropy between a pair of facing resin substrates (not illustrated). A conductive film (not illustrated) and an electrode (not illustrated) electrically connected to the conductive film are provided on the main surface of each resin substrate. Then, the light control film is driven by applying a voltage between the pair of conductive layers via the electrode.

[0098] The resin substrate is made of, for example, a transparent resin. The resin substrate may have, for example, polyethylene terephthalate (PET), polycarbonate (PC), or cycloolefin polymer (COP). In addition, for example, the above-described resins may be used in combination as the pair of facing resin substrates. The thickness of the resin substrate is, for example, in the range of 5 μm to 500 μm, preferably in the range of 10 μm to 200 μm, and more preferably in the range of 50 μm to 150 μm.

[0099] The conductive film may include, for example, a transparent conductive oxide, a transparent conductive polymer, a laminated film of a metal layer and a dielectric layer, a silver nanowire, a metal mesh of silver or copper, and the like. The thickness of the conductive film may be, for example, in a range of 200 nm to 2 μm.

[0100] Examples of the molecule having optical anisotropy include liquid crystals. That is, for example, a liquid crystal layer may be used as the molecular layer having optical anisotropy. Examples of the liquid crystal layer include polymer dispersed liquid crystal (PDLC), polymer network liquid crystal (PNLC), and guest-host liquid crystal. Alternatively, iodine or the like may be used as the molecule having optical anisotropy. The light control film may have a suspended particle device (SPD) including such a molecular layer.

[0101] In this example, the length of the antenna **30** in the direction along the outer edge (in this example, the upper edge **1***a*) of the glass plate **1** is longer than the length in the direction orthogonal to the direction along the outer edge. In this case, when the above-described line **33** (FIGS. **8** and **9**) extending from the antenna toward the outer edge is used, signal feeding to the antenna **30** becomes easy.

[0102] Although not particularly illustrated, the length of the antenna **30** in the direction along the outer edge of the glass plate **1** may be shorter than the length in the direction orthogonal to the direction along the outer edge. In this case, since the antenna **30** is easily brought close to the outer edge, the signal feeding to the antenna **30** is facilitated by the feeding structure of FIG. **7** described above.

[0103] The window glass for a vehicle **301** may include a light shielding layer **7** that shields visible light. The light shielding layer **7** is provided along the outer edge of the glass plate **1**. When the antenna **30** overlaps at least a part of the light shielding layer **7** in a plan view of the glass plate **1**, a portion overlapping the light shielding layer **7** is hardly visually recognized, so that the appearance of the window glass for a vehicle **301** is improved.

[0104] FIG. **11** is a plan view illustrating a specific example of the planar antenna according to the present embodiment. An antenna **30**A is an example of the antenna **30**. The antenna **30** includes a flat antenna element pattern **120** in which a slot **110** is formed. The slot **110** is an elongated notch formed in the antenna element pattern **120**.

[0105] The antenna element pattern **120** is an example of a film-like or plate-like flat conductor, and in this example, is a conductive film (film having conductivity) whose outer shape is formed in a substantially rectangular shape as a whole. The antenna element pattern **120** has an outer edge **191** on the first direction side, an outer edge **192** on the second direction side, an outer edge **193** on the third direction side, and an outer edge **194** on the fourth direction side.

[0106] The antenna element pattern **120** includes a flat first antenna element pattern **121** extending to one side with respect to the slot **110** and a flat second antenna element pattern **122** extending to the other side with respect to the slot **110**. In the present embodiment, first antenna element pattern **121** and second antenna element pattern **122** are separated by the slot **110**.

[0107] The first antenna element pattern **121** has a feeding area **103** to which a signal line (not illustrated) is electrically connected, and the second antenna element pattern **122** has a ground area **104** to which a ground line (not illustrated) is electrically connected. The feeding area **103** and the ground area **104** form a feeding portion of the antenna **30**A.

[0108] The slot **110** includes a slot **111**, a slot **112**, and a slot **113**. The slot **111**, the slot **112**, and the slot **113** are continuously connected in this connection order.

[0109] The slot **111** extends in a direction from the outer edge **194** toward the outer edge **192**. The slot **112** extends from an end in an extending direction of the slot **111** in a direction from the outer edge **191** toward the outer edge **193**. The slot **113** extends in a J shape from the end in the extending direction of the slot **112** to the outer edge **191** and opens at the outer edge **191**. The slot width at the open end of the slot **113** is wider than the slot width at the end in the extending direction of the slot **112**.

[0110] FIG. **12** is a diagram illustrating an example of a measurement result of the frequency characteristics of the antenna **30**A illustrated in FIG. **11**. FIG. **12** illustrates a measurement result of the antenna gain on the horizontal plane of the antenna **30**A in a state where the window glass for a vehicle in which the conductive member **100** is interposed between the glass plate and the antenna **30**A is attached to the vehicle.

[0111] In a case where the conductive member **100** is interposed, according to FIG. **12**, in a low frequency band of less than 3.3 GHZ, the antenna gain decreases due to the influence of the capacitance existing between the antenna **30**A and the conductive member **100**. On the other hand, in a high frequency band of 3.3 GHZ or more, a decrease in the antenna gain is suppressed, and a sufficient antenna gain functioning as an antenna is secured.

[0112] As described above, the embodiment has been described, but the above embodiment is presented as an example, and the present invention is not limited by the above embodiment. The above-described embodiments can be implemented in various other forms, and various combinations, omissions, substitutions, changes, and the like can be made without departing from the gist of the invention. These embodiments and modifications thereof are included in the scope and gist of the invention, and are included in the invention described in the claims and the equivalent scope thereof.

[0113] The present international application claims priority based on Japanese Patent Application No. 2022-186874 filed on Nov. 22, 2022, and the entire contents of Japanese Patent Application No. 2022-186874 are incorporated herein by reference.

[0114] With regard to the above embodiments, the following supplementary notes are further disclosed.

Supplementary Note 1

[0115] A window glass for a vehicle including

[0116] a laminated glass for a vehicle including a first glass plate having a first main surface and a second main surface opposite to the first main surface, a second glass plate having a third main surface facing the second main surface and a fourth main surface opposite to the third main surface, and an interlayer film disposed between the second main surface and the third main surface,

[0117] a planar antenna disposed between the second main surface and the third main surface, and

[0118] a conductive member that is separated from the antenna in a direction from the first glass plate toward the second glass plate and overlaps at least a part of the antenna in a plan view of the laminated glass.

Supplementary Note 2

[0119] The window glass for a vehicle according to Supplementary Note 1, in which the antenna is disposed between the second main surface and the interlayer film.

Supplementary Note 3

[0120] The window glass for a vehicle according to Supplementary Note 1 or 2, in which the conductive member is disposed between the interlayer film and the third main surface.

Supplementary Note 4

[0121] The window glass for a vehicle according to Supplementary Note 1 or 2, in which the interlayer film includes a first interlayer film and a second interlayer film sandwiching the conductive member in a thickness direction of the laminated glass.

Supplementary Note 5

[0122] The window glass for a vehicle according to Supplementary Note 1 or 3, in which the interlayer film includes a first interlayer film and a second interlayer film sandwiching the antenna in a thickness direction of the laminated glass.

Supplementary Note 6

[0123] The window glass for a vehicle according to Supplementary Note 5, in which

[0124] the second interlayer film is disposed between the first interlayer film and the third main surface, and

[0125] the conductive member is disposed between the second interlayer film and the third main surface.

Supplementary Note 7

[0126] The window glass for a vehicle according to any one of Supplementary Notes 1, 2, and 5, in which the conductive member is disposed on the fourth main surface.

Supplementary Note 8

[0127] The window glass for a vehicle according to any one of Supplementary Notes 1 to 7, in which the conductive member includes a conductive wire disposed at a predetermined interval and having a width narrower than the interval.

Supplementary Note 9

[0128] The window glass for a vehicle according to Supplementary Note 8, in which when a wavelength of a radio wave transmitted and received by the antenna with respect to a frequency F is λ, the interval is $\lambda/25$ or less.

Supplementary Note 10

[0129] The window glass for a vehicle according to any one of Supplementary Notes 1 to 9, in which the conductive member includes a planar conductive film.

Supplementary Note 11

[0130] The window glass for a vehicle according to Supplementary Note 10, in which the conductive film has a sheet resistance of 5 [Ω/□] or more.

Supplementary Note 12

[0131] The window glass for a vehicle according to Supplementary Note 10 or 11, in which the conductive film is a conductive film or a low-radiation film included in a light control film.

Supplementary Note 13

[0132] The window glass for a vehicle according to any one of Supplementary Notes 1 to 12, in which the conductive member includes a first bus bar, a second bus bar separated from the first bus bar, and a heating element connected between the first bus bar and the second bus bar.

Supplementary Note 14

[0133] The window glass for a vehicle according to Supplementary Note 13, in which

[0134] the first bus bar includes an upper portion extending in a direction along an upper edge of the laminated glass,

[0135] the second bus bar includes a lower portion extending in a direction along a lower edge of the laminated glass, and

[0136] the heating element includes a plurality of heating wires connected between the upper portion and the lower portion.

Supplementary Note 15

[0137] The window glass for a vehicle according to any one of Supplementary Notes 1 to 14, further including an electrode provided on the fourth main surface and electrically connected to the antenna.

Supplementary Note 16

[0138] The window glass for a vehicle according to Supplementary Note 15, in which when, in a plan view of the laminated glass, a region in which the conductive member is disposed is defined as a first region, and a region positioned outside the first region is defined as a second region, the electrode is positioned in the second region and faces the antenna or a line connected to the antenna on the same layer with the second glass plate interposed therebetween.

Supplementary Note 17

[0139] The window glass for a vehicle according to Supplementary Note 15 or 16, in which the electrode is connected to the antenna via a line passing through an outside of an end surface of the second glass plate.

Supplementary Note 18

[0140] The window glass for a vehicle according to any one of Supplementary Notes 1 to 17, in which in the antenna, a length in a direction along an outer edge of the laminated glass is longer than a length in a direction orthogonal to the direction along the outer edge.

Supplementary Note 19

[0141] The window glass for a vehicle according to any one of Supplementary Notes 1 to 18, further including a light shielding layer that is provided along the outer edge of the laminated glass and shields visible light,

[0142] in which the antenna overlaps at least a part of the light shielding layer in a plan view of the laminated glass.

[0143] From the disclosure thus described, it will be obvious that the embodiments of the disclosure may be varied in many ways. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure, and all such modifications as would be obvious to one skilled in the art are intended for inclusion within the scope of the following claims.

What is claimed is:

1. A window glass for a vehicle comprising:

a laminated glass for a vehicle including a first glass plate having a first main surface and a second main surface opposite to the first main surface, a second glass plate having a third main surface facing the second main surface and a fourth main surface opposite to the third main surface, and an interlayer film disposed between the second main surface and the third main surface;

a planar antenna disposed between the second main surface and the third main surface; and

a conductive member that is separated from the antenna in a direction from the first glass plate toward the second glass plate and overlaps at least a part of the antenna in a plan view of the laminated glass.

2. The window glass for a vehicle according to claim **1**, wherein the antenna is disposed between the second main surface and the interlayer film.

3. The window glass for a vehicle according to claim **2**, wherein the conductive member is disposed between the interlayer film and the third main surface.

4. The window glass for a vehicle according to claim **2**, wherein the interlayer film includes a first interlayer film and a second interlayer film sandwiching the conductive member in a thickness direction of the laminated glass.

5. The window glass for a vehicle according to claim **1**, wherein the interlayer film includes a first interlayer film and a second interlayer film sandwiching the antenna in a thickness direction of the laminated glass.

6. The window glass for a vehicle according to claim **5**, wherein

the second interlayer film is disposed between the first interlayer film and the third main surface, and

the conductive member is disposed between the second interlayer film and the third main surface.

7. The window glass for a vehicle according to claim **1**, wherein the conductive member is disposed on the fourth main surface.

8. The window glass for a vehicle according to claim **1**, wherein the conductive member is a conductive wire disposed at a predetermined interval and having a width narrower than the interval.

9. The window glass for a vehicle according to claim **8**, wherein when a wavelength of a radio wave transmitted and received by the antenna with respect to a frequency F is λ, the interval is $\lambda/25$ or less.

10. The window glass for a vehicle according to claim **1**, wherein the conductive member is a planar conductive film.

11. The window glass for a vehicle according to claim **10**, wherein the conductive film has a sheet resistance of 5 [Ω/□] or more.

12. The window glass for a vehicle according to claim **11**, wherein the conductive film is a conductive film or a low-radiation film included in a light control film.

13. The window glass for a vehicle according to claim **1**, wherein the conductive member includes a first bus bar, a second bus bar separated from the first bus bar, and a heating element connected between the first bus bar and the second bus bar.

14. The window glass for a vehicle according to claim **13**, wherein

the first bus bar includes an upper portion extending in a direction along an upper edge of the laminated glass,

the second bus bar includes a lower portion extending in a direction along a lower edge of the laminated glass, and the heating element includes a plurality of heating wires connected between the upper portion and the lower portion.

15. The window glass for a vehicle according to claim **1**, further comprising an electrode provided on the fourth main surface and electrically connected to the antenna.

16. The window glass for a vehicle according to claim **15**, wherein when, in a plan view of the laminated glass, a region in which the conductive member is disposed is defined as a first region, and a region positioned outside the first region is defined as a second region, the electrode is positioned in the second region and faces the antenna or a line connected to the antenna on the same layer with the second glass plate interposed therebetween.

17. The window glass for a vehicle according to claim **15**, wherein the electrode is connected to the antenna via a line passing through an outside of an end surface of the second glass plate.

18. The window glass for a vehicle according to claim **1**, wherein in the antenna, a length in a direction along an outer edge of the laminated glass is longer than a length in a direction orthogonal to the direction along the outer edge.

19. The window glass for a vehicle according to claim **18**, further comprising a light shielding layer that is provided along the outer edge of the laminated glass and shields visible light,

wherein the antenna overlaps at least a part of the light shielding layer in a plan view of the laminated glass.

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